

Question- Explain the operational differences between static and dynamic routing, and justify why configuring static routes would be the most efficient choice in this scenario.

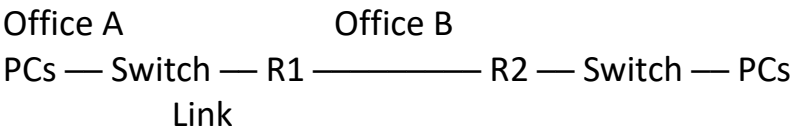
Static Routing vs Dynamic Routing

Feature	Static Routing	Dynamic Routing
Configuration	Manually configured by administrator	Learned automatically using routing protocols
Route changes	Administrator must update routes	Routes automatically adapt to changes
Protocols	No routing protocol required	RIP, OSPF, EIGRP, BGP, etc.
Resource usage	Very low CPU, RAM, and bandwidth usage	Uses CPU, RAM, and network bandwidth
Complexity	Simple for small networks	More complex to configure/manage
Failure recovery	Usually requires manual intervention	Can automatically find an alternative path
Best suited for	Small, stable networks	Large or frequently changing networks

Why Static Routing is better for this startup

The startup has **only two routers connecting two offices**, and the topology **rarely changes**. Therefore, static routing is the more efficient choice.

For example:



You could manually configure a route on **R1** pointing to Office B's network through R2, and a route on **R2** pointing back to Office A's network through R1.

For example:

- R1 → Office B network → R2
- R2 → Office A network → R1

Why this makes sense

- 1. Very small network** — Only two routers means there are very few routes to configure.

2. **Stable topology** — Since the network rarely changes, automatic route discovery isn't necessary.
3. **Less overhead** — No RIP/OSPF/EIGRP routing updates are needed.
4. **Simple management** — The administrator can easily understand and troubleshoot the routes.
5. **Predictable routing** — Traffic follows the manually defined path.
6. **No protocol configuration** — You don't need to configure a dynamic routing protocol.

Static routing is manually configured, while dynamic routing uses routing protocols to automatically learn and update routes. In this scenario, static routing is more efficient because there are only two routers and the topology rarely changes. Static routes require minimal resources, are simple to configure, and avoid the overhead of running a dynamic routing protocol. Dynamic routing would become more useful if the network grew or routes changed frequently.